THE MIDDLE FACTOR OF A COMPOSITE WHICH IS FLAT IS FLAT WHENEVER THE FIRST FACTOR IS SURJECTIVE AND FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The definition of flatness for morphisms of algebraic stacks through smooth covers by algebraic spaces. The author's section on flat morphisms, in which the counted statement occurs. Reduces the assertion to the corresponding assertion for morphisms of algebraic spaces.
- The corresponding permanence statement for flat morphisms of algebraic spaces. Invoked at the end of the counted proof (source lines 5448-5451) as a cross-chapter pointer. Supplies the statement to which the proof reduces.

Source reference. Original reference spaces-morphisms-lemma-flat-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5451–5451. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-flat-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X} \to \mathcal{Y} \to \mathcal{Z}$ be morphisms of algebraic stacks. If $\mathcal{X} \to \mathcal{Z}$ is flat and $\mathcal{X} \to \mathcal{Y}$ is surjective and flat, then $\mathcal{Y} \to \mathcal{Z}$ is flat.*

Proof. Choose an algebraic space W and a surjective smooth morphism $W \to \mathcal{Z}$. Choose an algebraic space V and a surjective smooth morphism $V \to W \times_{\mathcal{Z}} \mathcal{Y}$. Choose an algebraic space U and a surjective smooth morphism $U \to V \times_{\mathcal{Y}} \mathcal{X}$. We know that $U \to V$ is flat and that $U \to W$ is flat. Also, as $\mathcal{X} \to \mathcal{Y}$ is surjective we see that $U \to V$ is surjective (as a composition of surjective morphisms). Hence the lemma reduces to the case of morphisms of algebraic spaces. The case of morphisms

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of algebraic spaces is Morphisms of Spaces, Lemma spaces-morphisms-lemma-flat-permanence. □